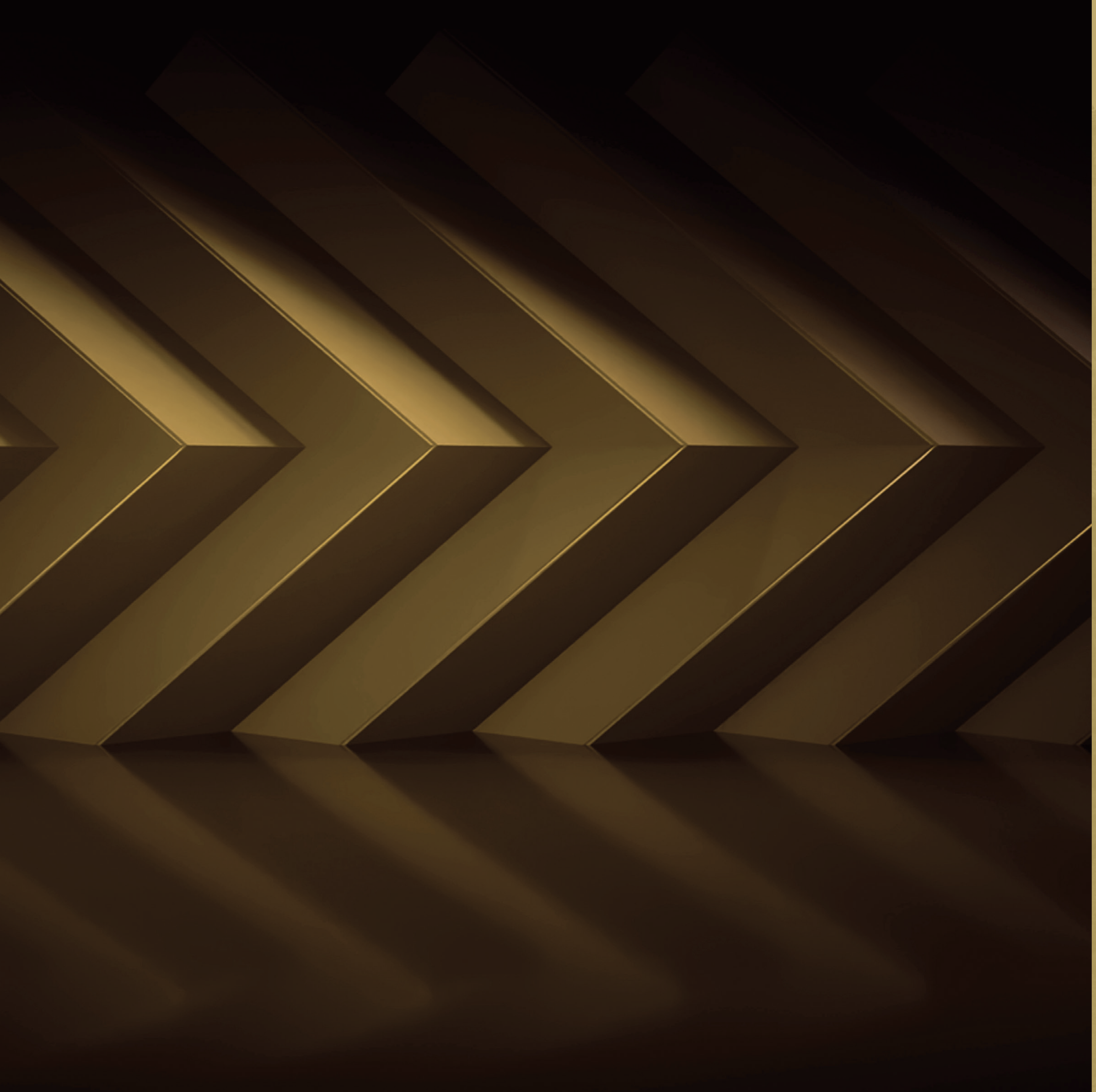
# THE JOURNEY TO EXASCALE

AMD × Hewlett Packard Enterprise

together we advance_



The Frontier and El Capitan exascale systems from the U.S. Department of Energy are leading us into a breakthrough era of efficiency in high-performance computing, artificial intelligence and a more sustainable, energy-efficient future.

# EXECUTIVE OVERVIEW

High-performance computing (HPC) has fundamentally changed the world of scientific research. Complex challenges that previously were unapproachable due to compute performance limitations and the massive power demands of legacy platforms can now be effectively addressed via powerful new advances in HPC.

The **U.S. Department of Energy** (DOE) is at the forefront of HPC, pushing its limits and advancing this technology into exascale compute performance levels that can effectively tackle the most complex challenges faced by the U.S. and the world.

These ambitious efforts led to the creation of **Frontier,** the first internationally recognized exascale system ever built for the DOE's **Oak Ridge National Laboratory,** and the even more powerful **El Capitan** for the DOE's **Lawrence Livermore National Laboratory.** Both systems—created in concert with **Advanced Micro Devices** (AMD) and **Hewlett Packard Enterprise** (HPE)—combine leading CPU, GPU and software technology into highly scalable compute platforms. These new supercomputers represent the beginning of HPC's next evolutionary phase that takes us into the exascale era.

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An exascale computer will perform one quintillion ($10^{18}$) operations per second.

## A BRIEF HISTORY OF HIGH-PERFORMANCE COMPUTING

Mainframes were the first massively scalable computer systems, but they were inflexible and difficult to program. This made them accessible only to a few organizations and limited their use to specific tasks. These limitations hindered innovation in the types of fast-paced, collaborative research that is now being executed in laboratories worldwide. HPC clusters overcame these accessibility and flexibility issues through new software and tools that empowered researchers.

However, as these systems were scaled up to offer greater functionality, they created their own set of challenges. Although thousands of lightning-fast nodes could be interconnected to create higher performing computing solutions, power became the limiting factor. For organizations to overcome these issues, a better and more efficient approach was needed to address compute and power issues simultaneously, enabling the achievement of exascale computing performance with limited power budgets.



## THE NEED FOR EXASCALE

Exascale is a measure of compute performance that is enumerated by a 1 in front of 18 zeros ($10^{18}$) and denotes the number of double-precision floating-point operations per second that a system can sustain. Exascale's massive performance capability would afford organizations like the DOE to more effectively address highly complex issues in energy, national and economic security, scientific discovery, manufacturing, earth and space science, healthcare, chemistry, physics, artificial intelligence (AI) and data analytics.

# A WORTHY JOURNEY

The journey to exascale has been long but highly productive and stimulating. In 1997, **ASCI Red** at **Sandia National Laboratory** achieved the first terascale performance ($10^{12}$). It took another 12 years for a system to break the petascale level ($10^{15}$) with **Los Alamos National Laboratory's Roadrunner** in 2008.

The DOE embarked on its exascale journey in 2012 by establishing a phased program to champion and attain exaflop-level performance. The first phase, *FastForward,* began with two research and development (R&D) investments (2012 and 2014) to lay the foundational groundwork. *DesignForward* (2013 and 2015) then focused on interconnect and system R&D. Finally, *PathForward* (2017) was the R&D project that focused on ensuring a stable supply chain by taking promising technologies for exascale computing and moving them closer to productization.

This collaborative strategy was established with AMD, HPE (Cray), IBM, Intel and NVIDIA. The DOE's investments in these programs have created a new level of cooperation among vendors and funded research into new products and technologies. This led to the development of **Frontier,** the first manifestation of exascale; **Aurora,** the second; and **El Capitan,** the world's first double-exascale system, providing additional performance and power efficiency benefits.

**TERASCALE ($10^{12}$)**

**1997**

## ASCI RED

Sandia National Laboratory
1.3 teraflops
850 kW

**PETASCALE ($10^{15}$)**

**2008**

## ROADRUNNER

Los Alamos National Laboratory
1.04 petaflops
2.35 MW

**2009**

## JAGUAR

Oak Ridge National Laboratory
2.3 petaflops
7 MW

**2012**

## TITAN

Oak Ridge National Laboratory
27 petaflops
9 MW

**2017**

## SUMMIT

Oak Ridge National Laboratory
200 petaflops
13 MW

**EXASCALE ($10^{18}$)**

**2021**

## FRONTIER

Oak Ridge National Laboratory
1.71 exaflops
24.6 MW

**2023**

## AURORA

Argonne National Laboratory
1.98 exaflops
38.7 MW

**2024**

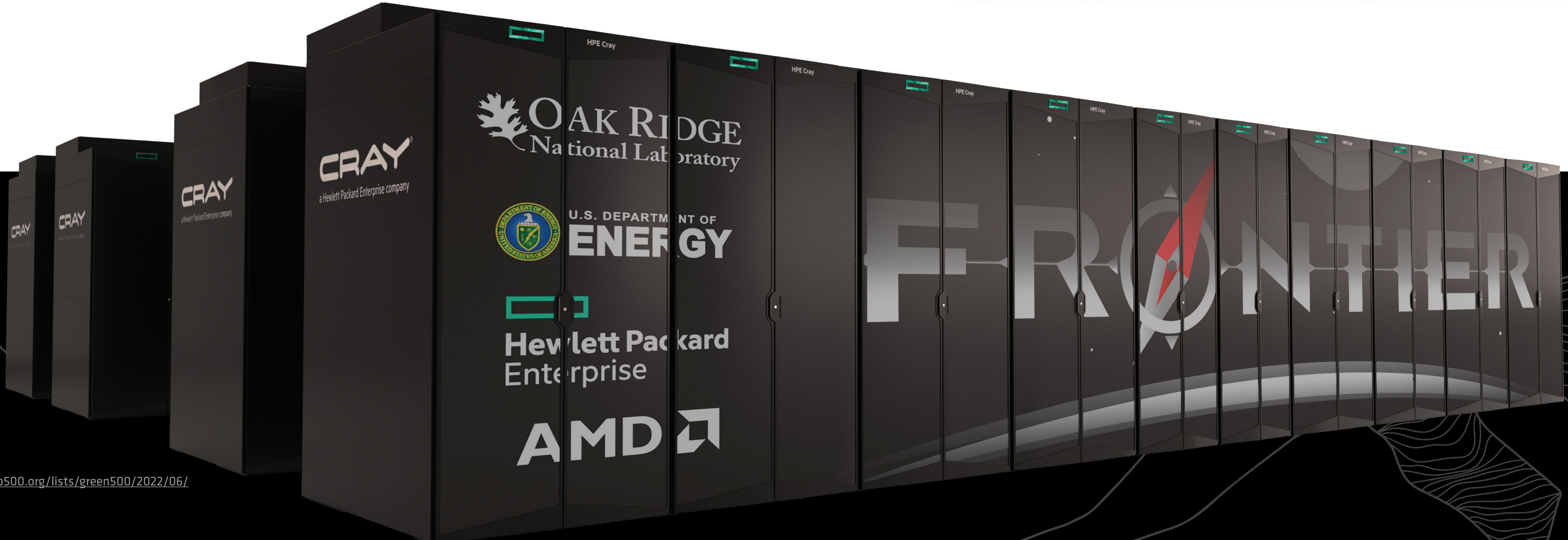
## EL CAPITAN

Lawrence Livermore National Laboratory
2.74 exaflops
29.5 MW

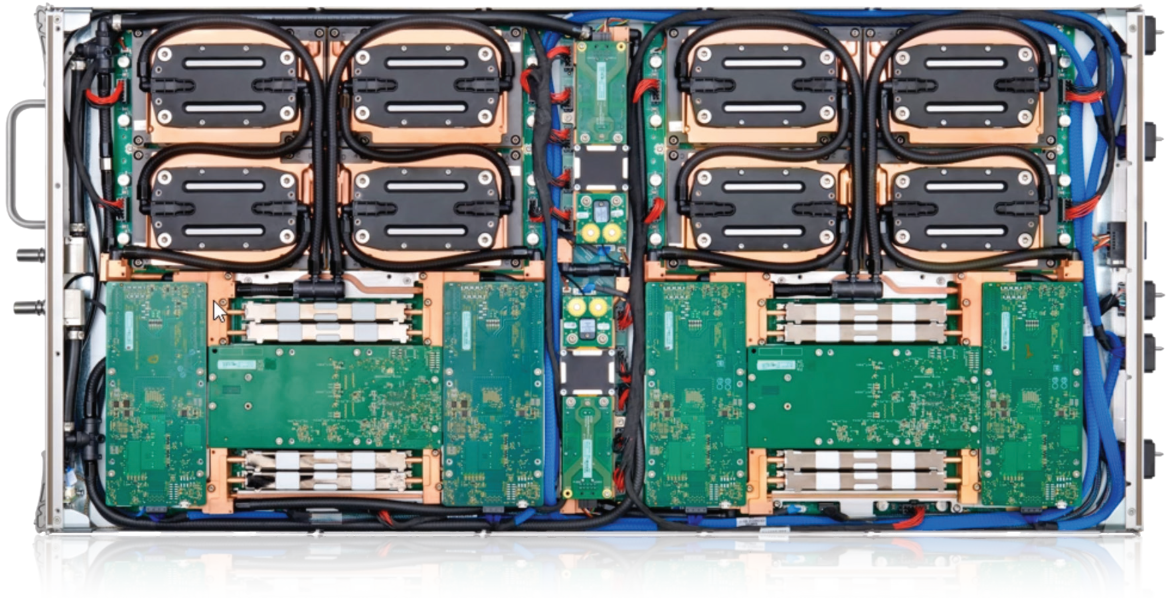
# FRONTIER

Built in 2021 at Oak Ridge National Laboratory in Oak Ridge, TN, Frontier was the world's first verified exascale system, achieving that level of performance in 2022. The system was based on the HPE Cray EX design that featured AMD CPU and GPU technologies. Occupying 74 19" rack cabinets, the system features 9,408 total compute nodes, along with storage and interconnect components. Frontier is based on HPE Cray EX235a blades with two compute nodes per blade, each node featuring one optimized AMD EPYC™ CPU and four AMD Instinct™ MI250X GPUs. The compute blades are connected via HPE Slingshot 11 64-port switches that provide 12.8 terabits/second of bandwidth.

A smaller Frontier test and development system (TDS) measured 62.68 gigaflops/watt, debuting on the Green500 list in 2022 as the world's most energy-efficient architecture at the time,[1] a nod to the power efficiency of the overall Frontier architecture.

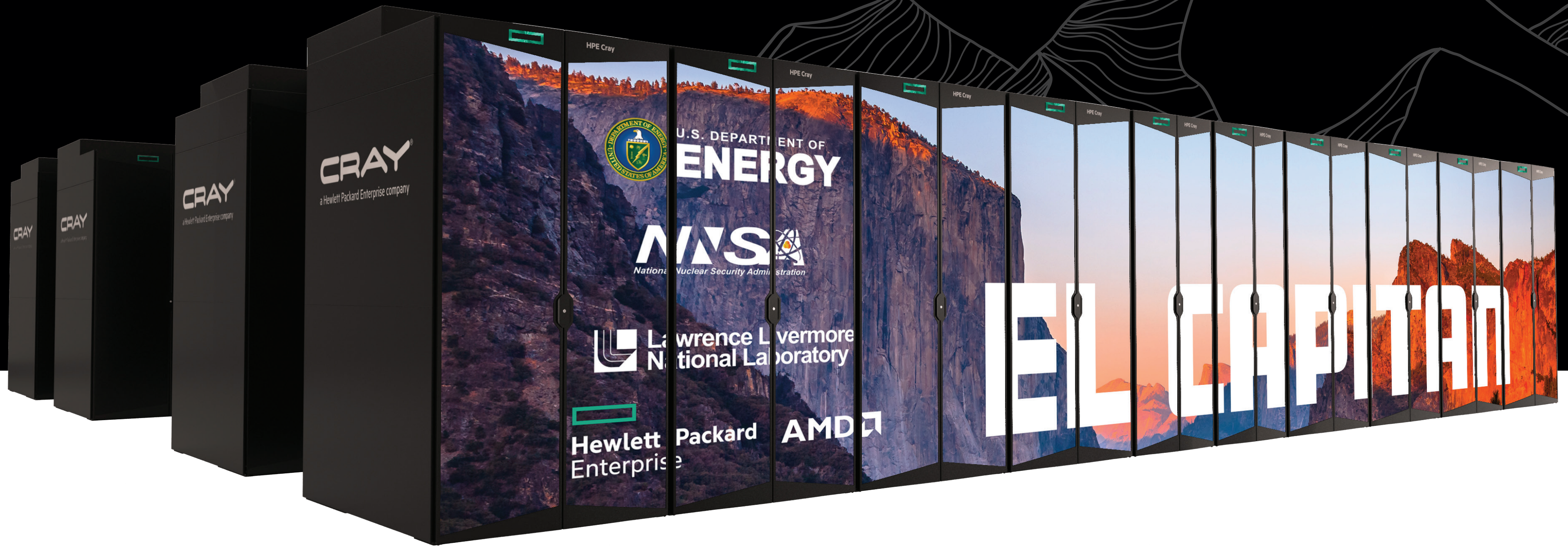



HPE Cray Supercomputing EX235a blade with AMD EPYC CPUs and AMD Instinct accelerators



1 Green500 June 2022 list, https://top500.org/lists/green500/2022/06/

In keeping with the DOE's power goals, peak power for the full system is 29.5 MW.



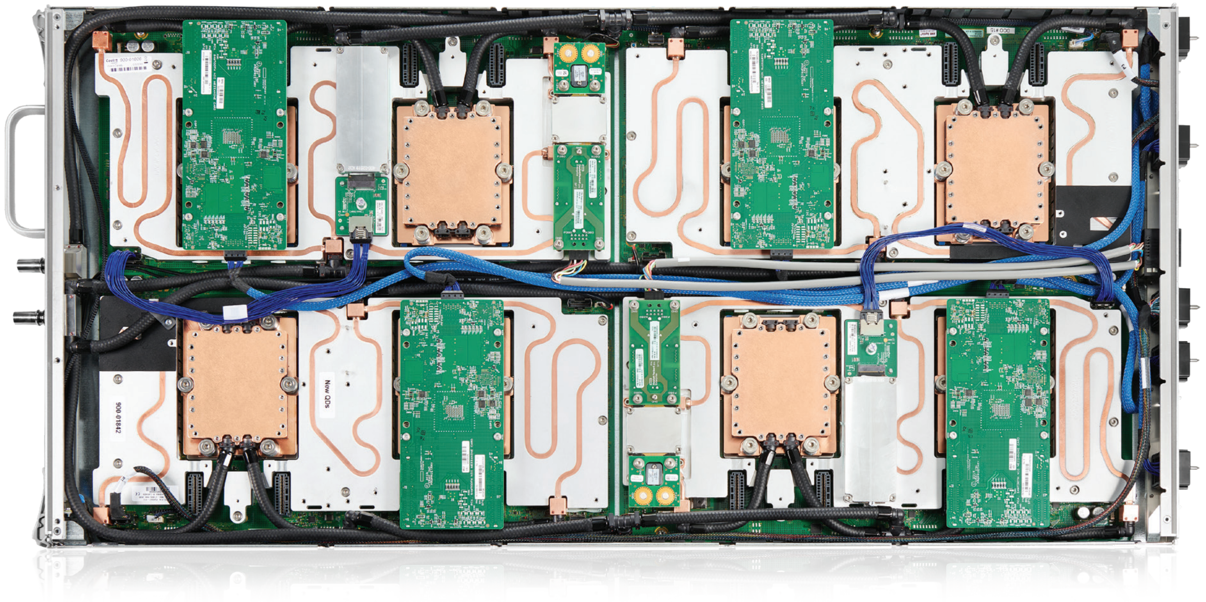

# EL CAPITAN

Using end-to-end HPE Cray EX systems that feature AMD technology, El Capitan was built at Lawrence Livermore National Laboratory in Livermore, CA. In November 2024, this system went into production, reaching 1.7 exaflops, making it the world's fastest supercomputer on the November 2024 TOP500 list.[2] Utilized by three national labs—Lawrence Livermore, Los Alamos and Sandia—El Capitan features an advanced design with HPE Cray EX255a blades whose two compute nodes are based on APUs (accelerated processing units) instead of distinct CPUs and GPUs. An APU packages CPU and GPU elements together; each AMD Instinct™ MI300A APU has three AMD "Zen 4" CPU chiplets (24 cores) and six AMD CDNA™ 3 (compute-centric) GPU Accelerated Compute Die (XCD) chiplets (228 compute units) integrated into a single package along with 128GB of high-bandwidth memory (HBM3) and high-speed I/O. The on-package advanced Infinity Architecture fabric unifies the CPU, GPU, HBM3 and I/O, providing high-speed communications between components.

The compute blades are connected via HPE Slingshot 11 64-port switches; however, because of the massive amount of compute performance that can be unleashed, maximum efficiency demands fast storage access. El Capitan overcame this challenge with a near-node local storage (NNLS) design that holds massive amounts of intermediate storage and is implemented to maximize system efficiency.

In keeping with the DOE's power goals, peak power for the full system, as recorded in the November 2024 Green500 list, rated El Capitan at 29.5 MW.[3]

HPE Cray Supercomputing EX255a blade with AMD Instinct APUs



2 TOP500 November 2024 list, https://top500.org/lists/top500/2024/11/; 3 Green500 November 2024 list, https://top500.org/lists/green500/list/2024/11/

# FOSTERING RESPONSIBLE ENERGY USE

Responsible energy consumption is a prime initiative for the DOE as it sets the tone for the country by ensuring that energy efficiency permeates all its activities. Therefore, while compute performance is essential, performance per watt is even *more* critical. To this end, the Green500 ranking, which measures power efficiency in computing, is as critical as the TOP500 ranking, which focuses on overall system performance. This dual goal is why there was a similarly intense focus on power efficiency in the Frontier system design, achieving the second highest spot for power efficiency in the June 2022 Green500 list with Frontier TDS, Frontier's development system, taking the top spot.[1]

Overall sustainability is also a critical design requirement, so the DOE, AMD and HPE took key steps to ensure these goals were met. The modularity/upgradeability of the HPE system, with its blade-forward design and end-to-end capabilities for optimal performance, scale and resiliency, is important as well, enabling the DOE to extend the value of the system over a longer time horizon if needed.

Systems that include discrete CPUs and GPUs (e.g. Frontier and previous systems) can handle CPU- or GPU-centric workloads through programming. Depending on the workload and the comparison point, APUs, which include CPU and GPU elements in a single package, have the potential to handle some workloads more efficiently, especially in relation to power efficiency.

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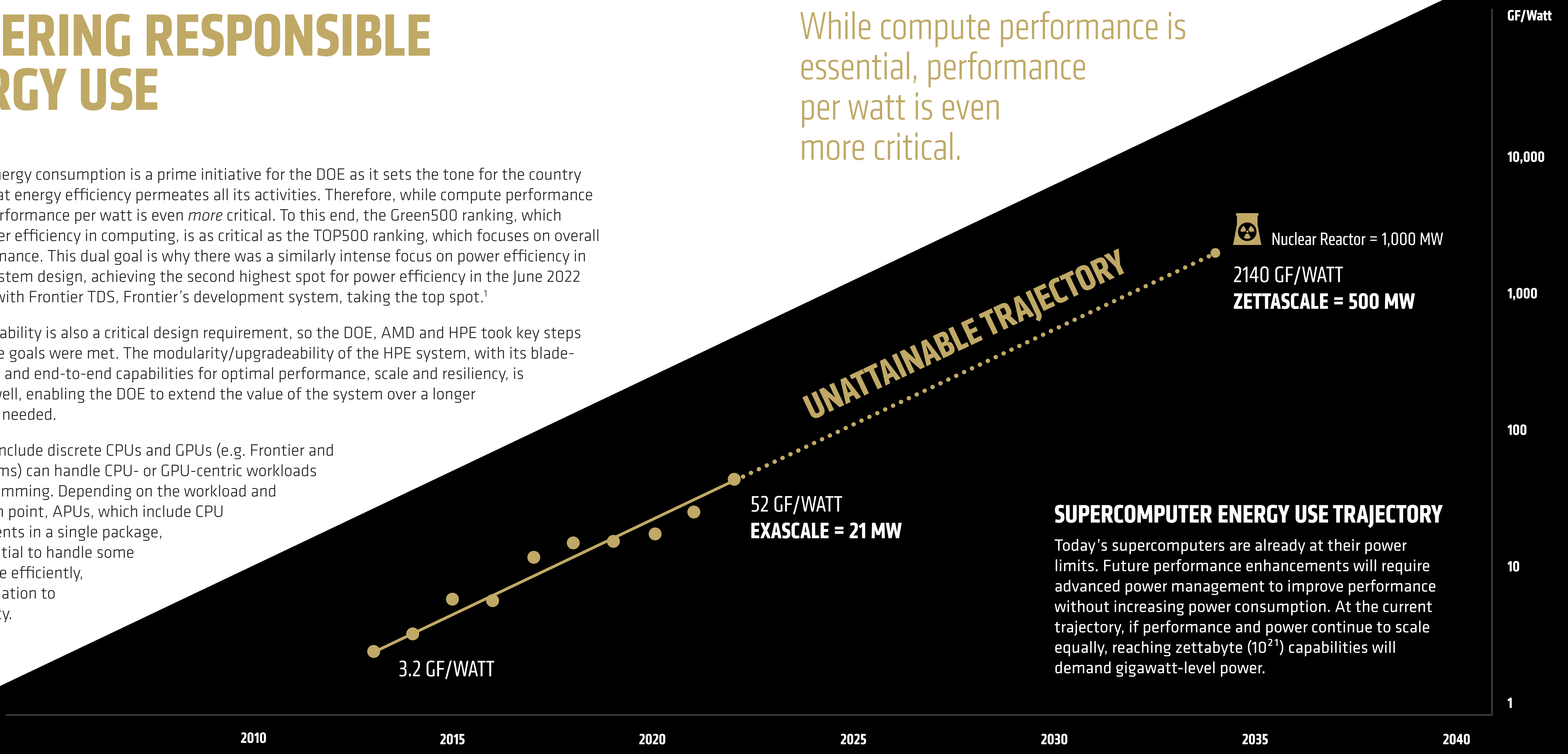



## SUPERCOMPUTER ENERGY USE TRAJECTORY

Today's supercomputers are already at their power limits. Future performance enhancements will require advanced power management to improve performance without increasing power consumption. At the current trajectory, if performance and power continue to scale equally, reaching zettabyte ($10^{21}$) capabilities will demand gigawatt-level power.



1 Green500 June 2022 list, https://top500.org/lists/green500/2022/06/

# WHAT EXASCALE MEANS TO THE WORLD

The dynamics of HPC are changing as exascale ushers in a new era of discovery and human progress. Researchers are just beginning to understand the capabilities of these systems, allowing them to re-evaluate some of the most significant problems facing humanity, such as climate change, with the enhanced computational power at their disposal. Over time, new and powerful HPC systems will further extend the capabilities that Frontier and El Capitan are delivering today. These will address more workloads, enable the potential for discovering new medical advances, answer questions about the universe or even apply AI to new research areas.

These systems have the potential to be a quantum leap forward for mankind. As compute performance has moved from teraflops to petaflops and now to exaflops, the world of research is being recast, opening new avenues previously unattainable and fueling even more growth for the future.

# INTO A NEW ERA, AND BEYOND

With its exascale strategy, the DOE aimed to deliver a flexible, extensible HPC solution tailored to its researchers' needs. The DOE, AMD and HPE collaborated to deliver Frontier, the DOE's first exascale HPC solution, driving significant performance increases. This solution empowers researchers to unlock the secrets of the universe and tackle thorny issues like global climate change. With the even more extraordinary capabilities of El Capitan, the DOE will continue taking on challenges that can only be addressed with massive computing capabilities as it not only eyes the world's most complex issues but also continues its work to further advance performance and power efficiency.

This ground-breaking program has paved the way for the U.S. to take the pole position in global HPC leadership and, in partnership with AMD and HPE, deftly steer the world into the future. Through continued strategic investments into HPC, especially solutions that make efficient use of power, the DOE is positioning itself to lead the exascale revolution and begin the groundwork for the next generation of performance to fuel tomorrow's research.

This ground-breaking program has cleared a path for the U.S. to take the pole position in global HPC leadership.

# A WORD FROM THE AMD CEO

“At AMD, our mission is to build great products that push the envelope in high-performance computing to help solve the world’s most important challenges. Frontier and El Capitan are great examples of our work, marking a historic milestone in supercomputing that is unlocking new possibilities in scientific research and pioneering breakthroughs across multiple fields. We are incredibly proud to contribute to this groundbreaking achievement and are focused on delivering future generations of AI compute that will significantly boost performance and initiate new types of research through ongoing public-private partnerships.”

**DR. LISA SU**, Chair and CEO, Advanced Micro Devices

# A WORD FROM THE HPE CEO

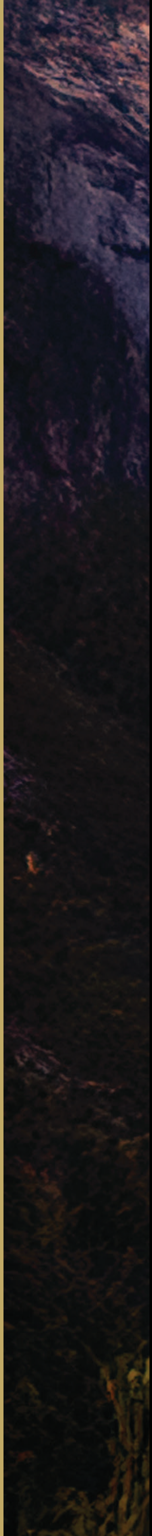
“Breaking the exascale speed barrier required achieving performance, energy efficiency and resiliency, which are all integral to system design. Alongside the U.S. Department of Energy (DOE), Oak Ridge National Laboratory and AMD, HPE applied more than 10 years of R&D to develop Frontier—the world’s first exascale-class supercomputer and one of the top energy-efficient systems specifically dedicated to open science and AI.

We are proud to have recently broken a new milestone in exascale, partnering with the DOE, National Nuclear Security Administration, Lawrence Livermore National Laboratory and AMD, on the launch of El Capitan, which delivered nearly 30% performance improvement over Frontier. El Capitan will answer some of the world’s most challenging problems, including unlocking new sources of renewable energy.”

**ANTONIO NERI**, President and CEO, Hewlett Packard Enterprise

Hewlett Packard Enterprise


THESE SYSTEMS HAVE
THE POTENTIAL TO
BE A QUANTUM LEAP
FORWARD FOR MANKIND.



# A SPECIAL THANK YOU

AMD and HPE thank the U.S. Department of Energy (DOE) for its leadership in this space. Reaching the lofty exascale goals was only possible with the vision, perseverance and collaboration the DOE provided to AMD and HPE throughout this journey. Additionally, the advances in software development required to take advantage of these systems would not have been possible without the guidance and commitment offered by the DOE throughout the past decade.

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